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Identifiability of Normal and Normal Mixture Models with Nonignorable Missing Data

Wang Miao^a, Peng Ding^b, and Zhi Geng^c

^aBeijing International Center for Mathematical Research, Peking University, Beijing, China; ^bDepartment of Statistics, University of California at Berkeley, Berkeley, CA, USA; ^cSchool of Mathematical Sciences and Center for Statistical Science, Peking University, Beijing, China

ABSTRACT

Missing data problems arise in many applied research studies. They may jeopardize statistical inference of the model of interest, if the missing mechanism is nonignorable, that is, the missing mechanism depends on the missing values themselves even conditional on the observed data. With a nonignorable missing mechanism, the model of interest is often not identifiable without imposing further assumptions. We find that even if the missing mechanism has a known parametric form, the model is not identifiable without specifying a parametric outcome distribution. Although it is fundamental for valid statistical inference, identifiability under nonignorable missing mechanisms is not established for many commonly used models. In this article, we first demonstrate identifiability of the normal distribution under monotone missing mechanisms. We then extend it to the normal mixture and t mixture models with nonmonotone missing mechanisms. We discover that models under the Logistic missing mechanism are less identifiable than those under the Probit missing mechanism. We give necessary and sufficient conditions for identifiability of models under the Logistic missing mechanism, which sometimes can be checked in real data analysis. We illustrate our methods using a series of simulations, and apply them to a real-life dataset. Supplementary materials for this article are available online.

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1. Introduction

Missing data arise in many biomedical and socioeconomic studies. In the presence of missing data, the observed data may not be representative for the population of interest, especially when the missing mechanism depends on the missing values themselves. For instance, if rich people tend not to respond in a survey, then the average wage obtained from the observed data will be lower than the truth.

An effective way to overcome this problem is to model the missing mechanism conditional on the observed covariates and the outcome. Using the terminologies in Rubin (1976), the missing mechanism is called missing at random (MAR) if it does not depend on the missing values themselves conditional on the observed data, and it is called missing not at random (MNAR) otherwise. In the current literature, a variety of estimation methodologies based on the MAR assumption have been proposed, including likelihood-based inference, imputation, inverse probability weighting, and doubly robust methods. However, MNAR is often the case in practice, when the missingness depends on the missing values even conditional on the observed covariates. Unfortunately, statistical inference becomes quite challenging with data subject to MNAR mechanisms, because the models are often not identifiable based on the observed data. Many authors have studied models under MNAR mechanisms. Among them, the most popular approach is to model the conditional distribution of the missing indicator given the outcome and covariates, termed the selection model (Little and Rubin 2002). Greenlees, Reece, and Zieschang (1982)

proposed maximum likelihood estimators for survey data with missing values, based on a fully parametric Logistic MNAR mechanism. Qin, Leung, and Shao (2002) proposed an empirical likelihood estimation procedure for the case with nonparametric outcome model and parametric missing mechanism. Ma, Kim, and Genton (2013) studied the semiparametric case with a symmetric outcome distribution and a parametric missing mechanism. Although they have developed useful estimation methods, the identifiability of their models may not be guaranteed even if the missing mechanism is parametric. Rotnitzky, Robins, and Scharfstein (1998) and Scharfstein and Irizarry (2003) developed methods for conducting sensitivity analysis by assuming completely specified outcome-dependent terms in the missing mechanisms. In his influential article in econometrics, Heckman (1979) proposed the Heckman selection model consisting of an outcome equation and a selection equation indicating the latent variable for the missing mechanism. In the Heckman selection model, the missing mechanism is nonignorable when the bivariate normal error terms of the outcome equation and selection equation are correlated. As a fundamental problem, valid statistical inference of these methods relies on their identifiability. Unfortunately, identifiability does not always hold even in parametric models.

When an “instrumental variable” is available, that is, there exists a variable that is associated with the outcome variable but independent of the missingness conditional on the outcome, identifiability of some MNAR models can be achieved. For example, Chen (2001) showed identifiability of a subset of the

regression parameters with biased sampling data, Wang, Shao, and Kim (2014) showed identifiability of certain nonparametric models with parametric missing mechanism, and Chen, Geng, and Zhou (2009) demonstrated the identifiability for binary outcomes with nonignorable missing data. However, it is not often feasible to find such an instrumental variable, without which the identifiability is not guaranteed in general.

In this article, we focus on the identifiability of models under MNAR mechanisms. We illustrate with counterexamples the potential difficulty for achieving identifiability under nonparametric outcome models in Section 2. We prove identifiability of the normal model under the monotone missing mechanism in Section 3 without requiring any instrumental variables. We find that models under the frequently used Logistic missing mechanism are less identifiable than those under the Probit missing mechanisms, and give necessary and sufficient conditions for the identifiability of models under the Logistic missing mechanism. In Section 4, we extend the results to normal mixture and t mixture models, which are useful to accommodate more complex data features such as heavy-tailedness and multimodality. In Section 5, we propose a latent monotone missing mechanism and establish their identifiability. In Section 6, we evaluate the finite sample properties of the nonignorable missing data models via a series of simulations. The simulation results show advantages of normal mixture and t mixture models for fitting complex data. In Section 7, we detect a nonignorable missing mechanism and latent components of the outcome variable in analysis of a real-life data on ambulatory expenditure. We conclude in Section 8, and relegate all the technical details to the supplementary materials.

2. Potential Difficulty for Nonparametric Identification

Throughout the article, we let X denote completely observed covariates, Y denote the outcome variable, and R denote the missing indicator of Y with $R = 1$ if Y is observed and $R = 0$ otherwise. We use lower-case letters to denote realized values of the corresponding random variables, for example, y for a realized value of Y . Suppose the observed data are n independently and identically distributed samples, with some values of Y missing.

The observed data allow us to identify only the observed distribution $P(y, R = 1|x)$, which, however, does not suffice to determine the joint distribution $P(y, r|x)$ without additional assumptions. There are two equivalent ways to factorize the joint distribution:

$$P(y, r|x) = P(y|x)P(r|x, y) = P(r|x)P(y|x, r),$$

with the first one being called the selection model and the second one being called the pattern mixture model (Little and Rubin 2002). Analogously, the observed distribution permits the following two equivalent factorizations:

$$\begin{aligned} P(y, R = 1|x) &= P(y|x)P(R = 1|x, y) \\ &= P(R = 1|x)P(y|x, R = 1). \end{aligned}$$

In this article, we adopt the selection model factorization for our discussion. Fundamentally, we are interested in identifying the joint distribution $P(y, r|x)$ by the observed distribution

$P(y, R = 1|x)$. We say that the model is identifiable, if and only if the joint distribution $P(y, r|x)$ can be uniquely determined by the observed distribution $P(y, R = 1|x)$, or, equivalently, two models yielding the same observed distribution must have the same joint distribution.

Because of their flexibility, nonparametric models are often used in the missing data literature (e.g., Qin, Leung, and Shao 2002; Ma, Kim, and Genton 2013). However, in general, the outcome distribution is not identifiable without specifying a parametric form for it, even if the missing mechanism is parametric. Below we provide three counterexamples to illustrate this potential difficulty.

Example 1. Consider the following models:

Model 1.1. $Y \sim \text{Unif}(-0.5, 0.5)$, and $P(R = 1|y) = \Phi(y)$, where $\Phi(y)$ is the distribution function of the standard normal distribution;

Model 1.2. Y has density $2\Phi(y)I\{-0.5 \leq y \leq 0.5\}$, and $P(R = 1|y) = \Phi(0) = 1/2$.

The models above have the same parametric Probit missing mechanism, but they have different outcome distributions. Nevertheless, they have the same observed distribution because

$$I\{-0.5 \leq y \leq 0.5\} \cdot \Phi(y) = 2\Phi(y)I\{-0.5 \leq y \leq 0.5\} \cdot 1/2.$$

Therefore, if we assume a nonparametric model for the outcome with a Probit missing mechanism, they cannot be distinguished by the observed data $P(y, R = 1)$.

Example 2. Consider the following models:

Model 2.1. $Y \sim \text{Exp}(2)$, $\text{logit } P(R = 1|y) = -\log 2 + y$;

Model 2.2. $Y \sim \text{Exp}(1)$, $\text{logit } P(R = 1|y) = \log 2 - y$.

The models above have the same parametric Logistic missing mechanism, but they have different outcome distributions. They have the same observed distribution because

$$2e^{-2y} \cdot \frac{e^{-\log 2 + y}}{1 + e^{-\log 2 + y}} = e^{-y} \cdot \frac{e^{\log 2 - y}}{1 + e^{\log 2 - y}}.$$

Therefore, they cannot be distinguished by the observed distribution.

The above counterexamples demonstrate the difficulty for obtaining identifiability of nonparametric outcome models even with parametric missing mechanisms. Unfortunately, even if we further restrict the outcome distributions to be symmetric as Ma, Kim, and Genton (2013) and assume parametric missing mechanisms, we still cannot identify the mean or distribution of the outcome.

Example 3. Consider the following models:

Model 3.1. $Y \sim N(1, 1)$, $\text{logit } P(R = 1|y) = -3/2 + y$;

Model 3.2. $Y \sim N(2, 1)$, $\text{logit } P(R = 1|y) = 3/2 - y$.

The models above have the same parametric Logistic missing mechanism, and symmetric outcome distributions. However, they cannot be identified by the observed distribution because

$$\phi(y - 1) \cdot \frac{\exp(-3/2 + y)}{1 + \exp(-3/2 + y)} = \phi(y - 2) \cdot \frac{\exp(3/2 - y)}{1 + \exp(3/2 - y)},$$

where $\phi(y)$ is the density of the standard normal distribution.

The above counterexamples show that nonparametric or semiparametric outcome models are generally not identifiable, even though the missing mechanisms are parametric. Without

identifiability, the estimates for the parameters of the nonparametric outcome models may be misleading and of limited interest in practice.

3. Normal Model with Nonignorable Missing Data

With nonignorable missing mechanisms, although identifiability of nonparametric outcome models is often hard to achieve as illustrated in Section 2, parametric outcome models are more likely to be identified. However, identifiability of many commonly used parametric models has not been established in the literature. In this section, we discuss identifiability of the normal model with nonignorable missing data. We first show the conditions for its identifiability without covariates, and illustrate the nonidentifiability for some nonignorable missing mechanisms, such as the commonly used Logistic missing mechanism. We then extend the result to nonignorable missing mechanisms with covariates, and use covariates to improve identifiability.

3.1. Identifiability without Covariates

Suppose $Y \sim N(\mu, \sigma^2)$, with the following missing mechanism:

$$P(R = 1|y) = F(\alpha + \beta y), \quad (1)$$

where $F(\cdot)$ is a known and strictly monotone distribution function with support on $(-\infty, +\infty)$. For instance, the standard normal distribution corresponds to the Probit missing mechanism, and the Logistic distribution corresponds to the Logistic missing mechanism. The missing mechanism is MAR if $\beta = 0$, and it is MNAR if $\beta \neq 0$. Mechanism (1) depicts that the response probability is monotone in the value of the outcome. For example, in some sensitive questionnaires, people with higher outcome values tend not to respond, and therefore $\beta < 0$.

The following condition about the tail behavior of $F(\cdot)$ plays a central role in our discussion.

Condition 1. For any $\delta > 0$, $\lim_{z \rightarrow -\infty} F(z)/e^{\delta z} = 0$ or $+\infty$.

Condition 1 requires the left tail decay rate of the response probability be not exponential. We can verify that the Probit missing mechanism satisfies **Condition 1**, but the Logistic missing mechanism does not because $\lim_{z \rightarrow -\infty} \{e^z/(1 + e^z)\}/e^z = 1$.

Theorem 1. Suppose $Y \sim N(\mu, \sigma^2)$ with missing mechanism (1). Then

- (a) σ^2 and $|\beta|$ are identifiable;
- (b) μ, σ^2, α , and β are identifiable, if the sign of β is known;
- (c) μ, σ^2, α , and β are identifiable, if **Condition 1** holds.

Result (a) above indicates that the variance of the outcome and the absolute value of its impact on the response probability are always identifiable, but the sign of its impact may not be identifiable. Result (b) implies that all parameters are identifiable if we have prior knowledge about the tendency of the missingness, that is, the sign of β . Result (c) shows that all parameters are identifiable, if $F(\cdot)$ satisfies **Condition 1**. **Example 3** in

Section 2 is an example illustrating that the normal model under the Logistic missing mechanism may not be identifiable. A similar example is also presented by Wang, Shao, and Kim (2014).

If $F(\cdot) = T_\nu(\cdot)$, the standard t distribution function with degrees of freedom ν , then it satisfies **Condition 1** when ν is known, and thus all parameters are identifiable. The model with $F(\cdot) = T_\nu(\cdot)$ for the missing mechanism is sometimes referred to as the Robit model (Liu 2004). By varying the degrees of freedom ν , the Robit models can be used to approximate many missing mechanisms. For instance, when ν tends to infinity, it approximates the Probit model; when ν is near 7 or 8, it approximates the Logistic model (Mudholkar and George 1978; Liu 2004). In fact, we have a stronger conclusion for the Robit model that the degrees of freedom parameter is also identifiable even if it is unknown.

Corollary 1. If $Y \sim N(\mu, \sigma^2)$, $P(R = 1|y) = T_\nu(\alpha + \beta y)$, and $\beta \neq 0$, then $\mu, \sigma^2, \nu, \alpha$, and β are all identifiable.

3.2. Identifiability with Covariates

When some completely observed covariates X are available, we assume the outcome model

$$Y|x \sim N\{\mu(x, \gamma), \sigma^2(x, \theta)\}, \quad (2)$$

and a generalized additive missing mechanism

$$P(R = 1|x, y) = F\{g(x, \alpha) + \beta y\}, \quad (3)$$

where $F(\cdot)$ is a known and strictly monotone distribution function, $\mu(\cdot, \cdot)$ and $\sigma(\cdot, \cdot)$ have known forms, and $(\gamma, \theta, \alpha, \beta)$ are unknown parameters. Note that (γ, θ, α) may be vectors. And we require that $\mu(\cdot, \gamma)$, $\sigma(\cdot, \theta)$, and $g(\cdot, \alpha)$ have a one-to-one mapping to γ, θ , and α , respectively. For instance, the linear function $\mu(x, \gamma) = x^T \gamma$ satisfies this condition.

Theorem 2. Assume that the outcome model is (2) and the missing mechanism is (3). Then

- (a) θ and $|\beta|$ are identifiable;
- (b) γ, θ, α , and β are identifiable, if the sign of β is known;
- (c) γ, θ, α , and β are identifiable, if **Condition 1** holds;
- (d) γ, θ, α , and β are identifiable, if the function pair $\{\mu(\cdot, \gamma), \sigma^2(\cdot, \theta)\}$ and the function $g(\cdot, \alpha)$ are linearly uncorrelated, that is, $a \cdot \mu(\cdot, \gamma) + b \cdot \sigma^2(\cdot, \theta) + g(\cdot, \alpha) \neq c$ for nonzero vector (a, b, c) and for all (γ, θ, α) .

In the above, (a), (b), and (c) are parallel with the results of **Theorem 1**. Result (d) gives another sufficient condition for improving identifiability by observed covariates X , when **Condition 1** does not hold. Results (d) does not allow the function $g(\cdot, \alpha)$ in the missing mechanism to be linearly correlated with the function pair $\{\mu(\cdot, \gamma), \sigma^2(\cdot, \theta)\}$, but it allows for linear dependence between the mean function $\mu(\cdot, \gamma)$ and the variance function $\sigma^2(\cdot, \theta)$. We illustrate **Theorem 2** by the following three examples.

Example 4. Assume that

$$Y|x \sim N(\gamma_0 + \gamma_1 x, \sigma^2), \quad P(R = 1|x, y) = \Phi(\alpha_0 + \alpha_1 x + \beta y).$$

The Probit missing mechanism satisfies [Condition 1](#) in [Theorem 2](#), and thus this model is identifiable. We can verify that this model is equivalent to the Heckman selection model (Heckman 1979) up to different parameterizations.

Contrary to the Probit missing mechanism, the Logistic missing mechanism does not satisfy [Condition 1](#) as pointed out before. Many researchers (e.g., Greenlees, Reece, and Zieschang 1982; Glynn, Laird, and Rubin 1986; Rotnitzky and Robins 1997; Qin, Leung, and Shao 2002) use the Logistic missing mechanisms when the outcome is MNAR. However, it should be noted that the Logistic missing mechanism is not identifiable even when some completely observed covariates are available, as shown in the following example.

Example 5. Assume that

$$Y|x \sim N(\gamma_0 + \gamma_1 x, \sigma^2), \quad \text{logit } P(R = 1|x, y) = \alpha_0 + \alpha_1 x + \beta y.$$

Let $\sigma^2 = 1$ and $\gamma_1 = 0.5$. Then the two different sets of parameters, $(\gamma_0, \alpha_0, \alpha_1, \beta) = (0, -2, -1, 2)$ and $(2, 2, 1, -2)$, lead to the same observed distribution. Therefore, the model above with the Logistic missing mechanism is not identifiable.

Example 6. Assume that

$$Y|x \sim N(\gamma_0 + \gamma_1 x, \sigma^2), \\ P(R = 1|x, y) = F(\alpha_0 + \alpha_1 x + \alpha_2 x^2 + \beta y),$$

where $\alpha_2 \neq 0$. Without any assumptions on $F(\cdot)$, the missing mechanism satisfies the condition in [Theorem 2\(d\)](#), where X has a linear impact on Y but a quadratic impact on its missingness. Thus all parameters are identifiable, even though [Condition 1](#) of [Theorem 2](#) may not be satisfied. For instance, even if $F(\cdot)$ is the Logistic distribution function, the model is still identifiable with covariates X . Therefore, the nonidentifiability problems in [Example 3](#) and [5](#) can be alleviated.

Note that in [Example 5](#), although $(\gamma_0, \alpha_0, \alpha_1, \beta)$ are not identifiable, the value of γ_1 is unique, which means that the slope of the outcome model is identifiable, but the intercept is not. We will show in the following theorem that this holds in general.

Theorem 3. Assume that

$$Y|x \sim N\{\mu(x, \gamma), \sigma^2\}, \quad P(R = 1|x, y) = F\{g(x, \alpha) + \beta y\}.$$

Then the partial derivatives $\partial\mu(x, \gamma)/\partial x$ and $|\partial g(x, \alpha)/\partial x|$ are always identifiable.

Going back to the linear model $Y|x \sim N(\gamma_0 + \gamma_1 x, \sigma^2)$ in [Example 6](#), [Theorem 3](#) says that even if $\alpha_2 = 0$, the coefficients of X in the outcome model, γ_1 , are always identifiable, although the intercept γ_0 may not be identifiable. The conclusion of [Theorem 3](#) can be further strengthened if $\mu(x, \gamma)$ and $g(x, \alpha)$ are both linear.

Corollary 2. Assume that

$$Y|x \sim N(\gamma_0 + x^T \gamma_1, \sigma^2), \quad P(R = 1|x, y) = F(\alpha_0 + x^T \alpha_1 + \beta y).$$

Assume that [Condition 1](#) fails, that is, $\lim_{z \rightarrow -\infty} F(z)/e^{\delta z} = c$ for some $\delta > 0$ and some $c \in (0, +\infty)$. Define $\tau_1 = 2\delta(\alpha_0 + \beta\gamma_0) + \delta^2\sigma^2\beta^2 + 2\log(c)$ and $\tau_2 = \alpha_1 + \beta\gamma_1$. Then

- (a) for a general distribution function $F(\cdot)$, $\tau_1 \neq 0$ or $\tau_2 \neq 0$ is a sufficient condition for identifiability of $(\gamma_0, \gamma_1, \sigma^2, \alpha_0, \alpha_1)$ and β ;
- (b) if $F(\cdot)$ is the Logistic distribution function, $\tau_1 \neq 0$ or $\tau_2 \neq 0$ is a necessary and sufficient condition for identifiability of $(\gamma_0, \gamma_1, \sigma^2, \alpha_0, \alpha_1)$ and β .

The vector τ_2 in the corollary above can be interpreted as the “total impact” of covariates X on the missingness of the outcome Y . However, the scalar τ_1 does not enjoy an apparent interpretation. From [Corollary 2](#), the parameters are not identifiable only in a small subset of the parameter space, that is, $\tau_1 = 0$ and $\tau_2 = 0$. In fact, as shown in the proof of [Corollary 2](#), even if $\tau_1 = 0$ and $\tau_2 = 0$, there are at most two possible sets of parameters with the same observed distribution, and they must satisfy: $\gamma'_1 = \gamma_1$, $\beta' = -\beta$, and $\alpha'_1 = -\alpha_1$. In practice, we may select one of the parameter sets based on our domain knowledge. From these equations, we can see that the absolute values of the vector α_1 are identifiable, implying that the absolute values of the coefficients of the covariates on the response probability are identifiable. Moreover, we can see from the above discussion that the parameters τ_1 and τ_2 are always identifiable. Therefore, we can consistently estimate them and test whether they are equal to zero. As a result, we can test the necessary and sufficient condition in (b) for the identifiability of the models based on the observed data.

4. Normal and t Mixtures with Nonignorable Missing Data

In this section, we discuss the identifiability of the normal mixture and t mixture outcomes with nonignorable missing data. These mixture distributions are useful to model outcomes with multiple modes or heavy tails. We first consider the normal mixture model

$$Y \sim \sum_{k=1}^K \pi_k N(\mu_k, \sigma_k^2), \quad \sum_{k=1}^K \pi_k = 1, \quad \pi_k \geq 0, \\ k = 1, \dots, K. \quad (4)$$

The model above includes mixtures in both location and scale parameters. If $\mu_k = \mu$ for all k , it is a scale mixture; and if $\sigma_k = \sigma$ for all k , it is a location mixture. As in [Section 3.1](#), we assume that the missing mechanism is model (1), where $F(\cdot)$ is a strictly monotone distribution function. It satisfies some of [Condition 1](#) discussed before, and [Conditions 2](#) and [3](#) below.

Condition 2. For any $\theta_0 \in (-\infty, +\infty)$, $\theta_1 > 0$, $\delta_1, \delta_2 \geq 0$, and $\delta_1 + \delta_2 \neq 0$, the limit $\lim_{z \rightarrow +\infty} \{F(\theta_0 + \theta_1 z) - F(z)\}/e^{-\delta_1 z^2 - \delta_2 z} = 0$ or ∞ .

Condition 3. For any $\theta_0 \in (-\infty, \infty)$, $\theta_1 > 0$, and $M > 0$, at most one of $\lim_{z \rightarrow +\infty} z^M \{F(\theta_0 + \theta_1 z) - F(z)\}$ and $\lim_{z \rightarrow -\infty} z^M \{1 - F(\theta_0 + \theta_1 z)/F(z)\}$ is finite and positive.

[Condition 2](#) requires the tail behavior of $F(\theta_0 + \theta_1 z) - F(z)$ be different from any normal or exponential density. It is straightforward to verify that the commonly used Probit and Robit missing data mechanisms satisfy [Condition 2](#), but the

Logistic missing mechanism does not. [Condition 3](#) holds for all Logistic, Probit, and Robit missing mechanisms.

We have the following result on the identifiability of normal mixture models.

Theorem 4. Suppose that the outcome model is a normal mixture (4) with an unknown K and π_k 's, and the missing mechanism is (1) with a known $F(\cdot)$ satisfying [Conditions 1](#) and [2](#). Then all the parameters $K, \{(\pi_k, \mu_k, \sigma_k^2) : k = 1, \dots, K\}, \alpha$ and β are identifiable.

Next we discuss identifiability of the t model and its location mixture. A t random variable can be represented by an infinite scale mixture of normal random variables (Little and Rubin 2002; Liu 2004), and thus the mixture of t random variables can be viewed as an infinite location and scale mixture of normal random variables. It can accommodate multimodality and heavy-tailedness simultaneously, and we can tune the degrees of freedom and the number of components in the mixture distribution in practice. We consider the outcome model

$$Y \sim \sum_{k=1}^K \pi_k T_v(\mu_k, \omega^2), \quad \sum_{k=1}^K \pi_k = 1, \quad \pi_k \geq 0, \quad k = 1, \dots, K, \quad (5)$$

where $T_v(\mu_k, \omega^2)$ is a t random variable with location parameter μ_k , scale parameter ω^2 , and degrees of freedom v .

Theorem 5. Suppose that the outcome follows a location mixture of t distributions in (5) with unknown K, π_k 's, and v , and the missing mechanism is (1) with a known $F(\cdot)$ satisfying [Condition 3](#). Then all the parameters $K, \{(\pi_k, \mu_k) : k = 1, \dots, K\}, \omega^2, v, \alpha$, and β are identifiable.

As a special case of the t mixture with $K = 1$, the parameters are also identifiable if the outcome model follows a single t distribution.

5. Latent Monotone Missing Mechanism

In this section, we still assume that the outcome Y comes from a normal mixture distribution with K components indexed by parameters (μ_k, σ_k^2) and mixing proportions π_k for $k = 1, \dots, K$. Note that K and π_k 's may be unknown. We further allow the missing mechanism to depend on the indicator, G , of the latent components of the mixture model

$$P(R = 1|y, G = k) = F(\alpha_k + \beta_k y), \quad (6)$$

where $F(\cdot)$ is a known distribution function, α_k and β_k are unknown parameters depending on latent groups. Unfortunately, this general missing mechanism is lack of identifiability. Even for the case with $\beta_k = 0$, that is, a latent ignorable missing mechanism (Frangakis and Rubin 1999), the model is not identifiable as illustrated in the following example.

Example 7. Assume that

$$Y \sim \pi_1 N(0, 1) + (1 - \pi_1)N(0, 4), \\ P(R = 1|y, G = k) = F(\alpha_k) \quad (k = 1, 2).$$

Then any two models with $\pi_1 F(\alpha_1) = (1 - \pi_1)F(\alpha_2) = 1/4$ result in the same observed distribution.

Therefore, we should impose some restrictions on model (6). As a motivating example, wage levels vary among different cities or communities, and in a survey an individual will decide to respond or not based on the average wage level of his city or community. In other words, it is the wage deviation of an individual from the city's average level that influences the response probability directly. The variance of the wages, a measure of the gap between the rich and the poor people, may also influence the response probability. Based on this intuition, we assume the following latent monotone missing mechanism, which depends on the mean and variance of the latent group:

$$P(R = 1|y, G = k) = F\left[\frac{\alpha\psi(\sigma_k) + \beta\{y - \kappa(\mu_k)\}}{\varphi(\sigma_k)}\right], \quad (7)$$

where $\psi(\cdot), \kappa(\cdot)$, and $\varphi(\cdot)$ are known functions. We require that $\kappa(\cdot)$ and $\varphi(\cdot)$ be increasing, and $\psi(\cdot)$ and $\varphi(\cdot)$ be positive. The common parameters α and β represent similar missing data patterns in different latent groups. Given the component indicator of the mixture distribution, the missing mechanism (6) is monotone in the outcome. Averaged over the latent components, however, the missing mechanism may not be monotone.

Example 8. If $\psi(\sigma) = 1, \varphi(\sigma) = 1$ and $\kappa(\mu) = 0$, we have the monotone missing mechanism $P(R = 1|y, G = k) = F(\alpha + \beta y)$, as discussed in (1) and [Sections 3](#) and [4](#).

Example 9. If $\psi(\sigma) = \sigma, \varphi(\sigma) = \sigma$, and $\kappa(\mu) = \mu$, we have

$$P(R = 1|y, G = k) = F\left(\alpha + \beta \frac{y - \mu_k}{\sigma_k}\right).$$

The missing mechanism depends on the deviation from the center of each component $(y - \mu_k)/\sigma_k$, and the missing proportions are the same for all components.

Example 10. If $\psi(\sigma) = 1, \varphi(\sigma) = \sigma$, and $\kappa(\mu) = \mu$, we have

$$P(R = 1|y, G = k) = F\left(\frac{\alpha}{\sigma_k} + \beta \frac{y - \mu_k}{\sigma_k}\right).$$

The missing mechanism depends on the deviation from the center and the variance of each component, and the missing proportions are larger for components with larger variances if $\alpha < 0$. If $\mu_k = \mu$ for all components, then we have a scale mixture of normal distributions. As K increases, if the empirical distribution of σ_k approaches a scaled-inverse- χ^2 distribution, then the resulting marginal outcome distribution is a t distribution. We can verify that, under the above missing mechanism with $F(\cdot) = \Phi(\cdot)$, the model is equivalent to the selection- t model (Marchenko and Genton 2012).

Example 11. If $\psi(\sigma) = 1, \varphi(\sigma) = \sigma^2$, and $\kappa(\mu) = \mu$, we have

$$P(R = 1|y, G = k) = F\left(\alpha\sigma_k + \beta \frac{y - \mu_k}{\sigma_k}\right).$$

The missing mechanism depends on the deviation from the center and the variance of each component, and the missing proportions are smaller for components with larger variances if $\alpha < 0$.

In the above examples, we suggested several candidate missing mechanisms that are easy to interpret. Practitioners can choose their own missing mechanisms according to their background knowledge, as long as the functions $\{\psi(\cdot), \varphi(\cdot), \kappa(\cdot)\}$

satisfy the conditions above. Below we show the identifiability of model (4) under the missing mechanism (7).

Theorem 6. Suppose that the outcome model is a normal mixture (4) with unknown K and π_k 's, and the missing mechanism is (7). If $F(\cdot)$ is known and satisfies Conditions 1 and 2, then the parameters K , $\{(\pi_k, \mu_k, \sigma_k^2) : k = 1, \dots, K\}$, α and β are identifiable.

The missing mechanism (1) is a special case of the missing mechanism (7) as shown in Example 8, and therefore Theorem 4 can be viewed as a special case of Theorem 6. Similar to Section 2.2, we can incorporate covariates into model (7), and establish identifiability results for the parameters conditional on the covariates.

6. Simulation Studies

In this section, we use simulations to evaluate the finite sample properties of the models discussed in the previous sections. The data-generating models are identifiable according to the theorems and corollaries in the previous sections. For each model, we simulate 1000 independent datasets under sample sizes 500 and 1500, and summarize the results with boxplots.

6.1. Normal Outcomes

We generate the outcome variable $Y \sim N(0, 4)$, and choose Probit, Logit, and Robit (with $\nu = 2, 16$) missing mechanisms. We choose $\alpha = 1$ and $\beta = 1$ or 2 for the missing mechanisms. For these settings, the missing data proportions are between 30% and 40%. For each dataset, we apply various missing mechanisms to estimate the parameters, including Probit, Logistic, and Robit models with unknown degrees of freedom.

We show the simulation results only for $\beta = 2$ in Figure 1, and those for $\beta = 1$ have similar patterns and are omitted. From Figure 1, we can see that misspecification of the missing mechanism has little influence on estimation of the mean of the outcome, although it has some influence on the estimation of the missing mechanism itself. It is because the quantiles of normal, Logistic, and t distributions are almost linearly correlated over a large range (Mudholkar and George 1978). When the true missing mechanism is Robit with a small degrees of freedom and the Probit model is used for estimation, it leads to small biases for estimating μ , but large biases for estimating β . As the degrees of freedom increase, the biases for both μ and β become smaller, and they can be improved by increasing the sample size.

The Probit model has explicit form of observed likelihoods, but those of the Logistic model and the Robit model involve integrals. According to the simulation results, and for computational convenience, we recommend using the Probit missing mechanism for estimation of the mean of the outcome if we do not care about the missing mechanism itself. Otherwise, we recommend conducting sensitivity analysis using the Robit missing mechanism with different degrees of freedom.

6.2. Scale Mixtures of Normal and t Outcomes

We use two-component scale mixtures of normal and t distributions with $\nu = 5$ for the outcome model. We choose both

the Probit and latent Probit missing mechanisms. We generate a covariate $X_1 \sim N(1, 1)$, let $X = (1, X_1)^T$, and then generate outcomes from the following models:

1. Scale mixtures of normal outcome with the Probit missing mechanism

$$Y|x \sim \sum_{k=1}^2 \pi_k N(x^T \gamma, \sigma_k^2), \quad P(R = 1|x, y) = \Phi(x^T \alpha + \beta y);$$

2. Scale mixtures of normal outcome with the latent Probit missing mechanism

$$Y|x \sim \sum_{k=1}^2 \pi_k N(x^T \gamma, \sigma_k^2),$$

$$P(R = 1|x, y, G = k) = \Phi \left\{ \frac{x^T \alpha + \beta(y - x^T \gamma)}{\sigma_k} \right\};$$

3. t outcome with the Probit missing mechanism

$$Y|x, \sigma \sim N(x^T \gamma, \sigma^2), \quad \sigma^2 \sim \frac{\omega^2 \nu}{\chi_\nu^2},$$

$$P(R = 1|x, y) = \Phi(x^T \alpha + \beta y);$$

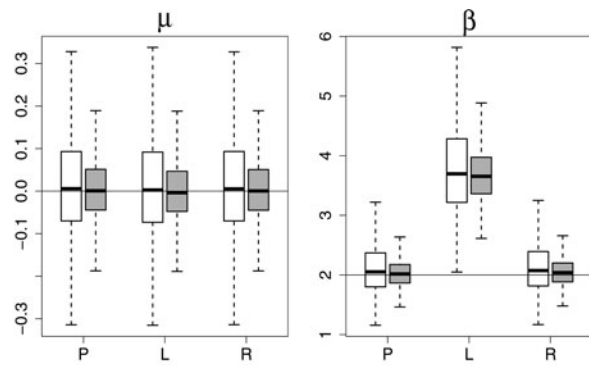
4. t outcome with the latent Probit missing mechanism (or, equivalently, the selection- t model)

$$Y|x, \sigma \sim N(x^T \gamma, \sigma^2), \quad \sigma^2 \sim \frac{\omega^2 \nu}{\chi_\nu^2},$$

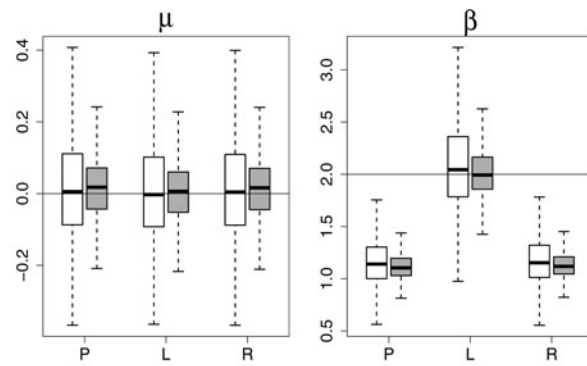
$$P(R = 1|x, y, \sigma) = \Phi \left\{ \frac{x^T \alpha + \beta(y - x^T \gamma)}{\sigma} \right\}. \quad (8)$$

The true parameters for the above cases 1 and 2 are $\pi_1 = \pi_2 = 0.5$, $\gamma = (1, 1)^T$, $\sigma_1^2 = 1$, $\sigma_2^2 = 4$, and $\alpha = (1, 1)^T$. The true parameters for the above cases 3 and 4 are $\nu = 5$, $\gamma = (1, -1)^T$, $\omega^2 = 1$, and $\alpha = (1, 1)^T$. Further, we set $\beta = -0.5$ and -1 , which result in about 15% and 25% missing values, respectively.

We fit the generated datasets by three models: scale mixture of normal outcomes with correct missing mechanisms, normal outcome model with Probit missing mechanism, and the selection- t model. We show the simulation results for $\beta = -0.5$ in Figure 2, and those for $\beta = -1$ are similar and omitted. To save space, we present only the results for the coefficients of X_1 and Y . From Figure 2, when the missing mechanism is Probit (cases 1 and 3), the estimators of the three methods have small biases for outcome model parameter γ , but the estimators of the normal outcome model and the selection- t model have large biases for missing mechanism parameters (α, β) . This is because the former model cannot accommodate heavy-tailedness of the outcome distribution and the latter is not a monotone missing mechanism. Using a scale mixture of normal outcome models, biases of both outcome model parameter γ and missing mechanism parameters (α, β) become smaller as sample size increases (from 500 to 1500), but they become even larger using the other two models. When the missing mechanism is a latent Probit model (cases 2 and 4), the selection- t models are the true specifications for case 4. Thus, they have smallest biases for γ , α , and β overall. But a scale mixture of normal models also works very well for case 4, and their biases for (γ, α, β) are close to those of the selection- t model. However,



(a) Generated from the Probit missing mechanism



(b) Generated from the Logistic missing mechanism

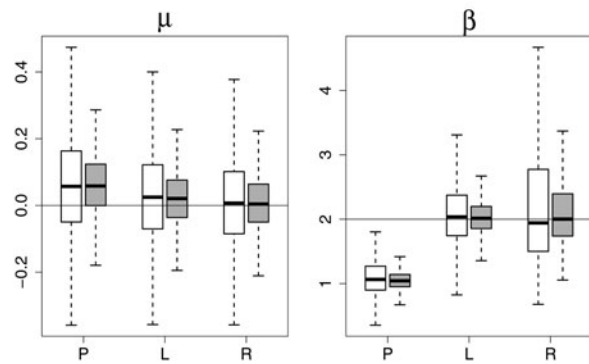
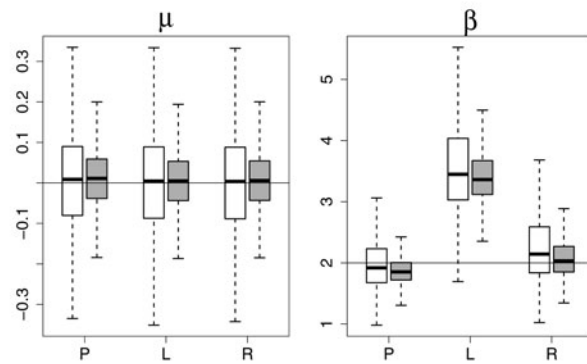
(c) Generated from the Robit missing mechanism ($\nu = 2$)(d) Generated from the Robit missing mechanism ($\nu = 16$)

Figure 1. Estimates when $\beta = 2$. Data are generated from normal outcome models with different missing mechanisms, and analyzed under normal outcome models with different missing mechanisms: Probit (P), Logistic (L), and Robit (R) with unknown degrees of freedom. In each boxplot, white boxes are for sample size 500 and gray boxes for 1500. The horizontal lines illustrate the true values of the parameters.

for case 2, the selection- t models have large biases for missing mechanism parameters (α , β), and they do not improve as sample size increases. Among these estimation methods, the normal outcome model is the worst. So we recommend a scale mixture of normal models for heavy-tailed outcomes because they enjoy robustness and easy interpretations.

6.3. General Mixtures of Normal and Location Mixtures of t Outcomes

We generate data from two-component general mixtures of normal (mixture in both location and scale) outcome models with both the Probit and latent Probit missing mechanisms; and two-component location mixture of t outcome models with the Probit missing mechanisms. We first generate a covariate $X_1 \sim N(1, 1)$, let $X = (1, X_1)^T$, and then generate the outcome according to the following models:

1. Normal mixture outcome with the Probit missing mechanism

$$Y|x \sim \sum_{k=1}^2 \pi_k N(x^T \gamma_k, \sigma_k^2),$$

$$P(R = 1|x, y) = \Phi(x^T \alpha + \beta y),$$

with true parameters $\pi_1 = \pi_2 = 0.5$, $\gamma_1 = (1, 1)^T$, $\gamma_2 = (1, -1)^T$, $\sigma_1^2 = 1$, $\sigma_2^2 = 4$, and $\alpha = (1, 1)^T$.

2. Normal mixture outcome with the latent Probit missing mechanism

$$Y|x \sim \sum_{k=1}^2 \pi_k N(x^T \gamma_k, \sigma_k^2),$$

$$P(R = 1|x, y, G = k) = \Phi \left\{ \frac{x^T \alpha + \beta(y - x^T \gamma_k)}{\sigma_k} \right\},$$

with true parameters $\pi_1 = \pi_2 = 0.5$, $\gamma_1 = (2, 1)^T$, $\gamma_2 = (-1.5, -2)^T$, $\sigma_1^2 = 1$, $\sigma_2^2 = 4$, and $\alpha = (1.5, -2)^T$.

3. Location mixtures of t outcome with the Probit missing mechanism

$$Y|x, \sigma \sim \sum_{k=1}^2 \pi_k N(x^T \gamma_k, \sigma^2), \quad \sigma^2 \sim \frac{\omega^2 \nu}{\chi_\nu^2},$$

$$P(R = 1|x, y) = \Phi(x^T \alpha + \beta y),$$

with true parameters $\nu = 5$, $\pi_1 = \pi_2 = 0.5$, $\gamma_1 = (1, 3)^T$, $\gamma_2 = (1, -3)^T$, $\omega^2 = 1$, and $\alpha = (1, 1)^T$.

We choose $\beta = -0.5$ and -1 for missing mechanisms. Under these settings, the missing data proportions are between 20% and 50%. Note that these three simulations are different from those in the previous subsection because the outcome models here allow for location mixtures, which usually lead to multimodality and have more parameters.

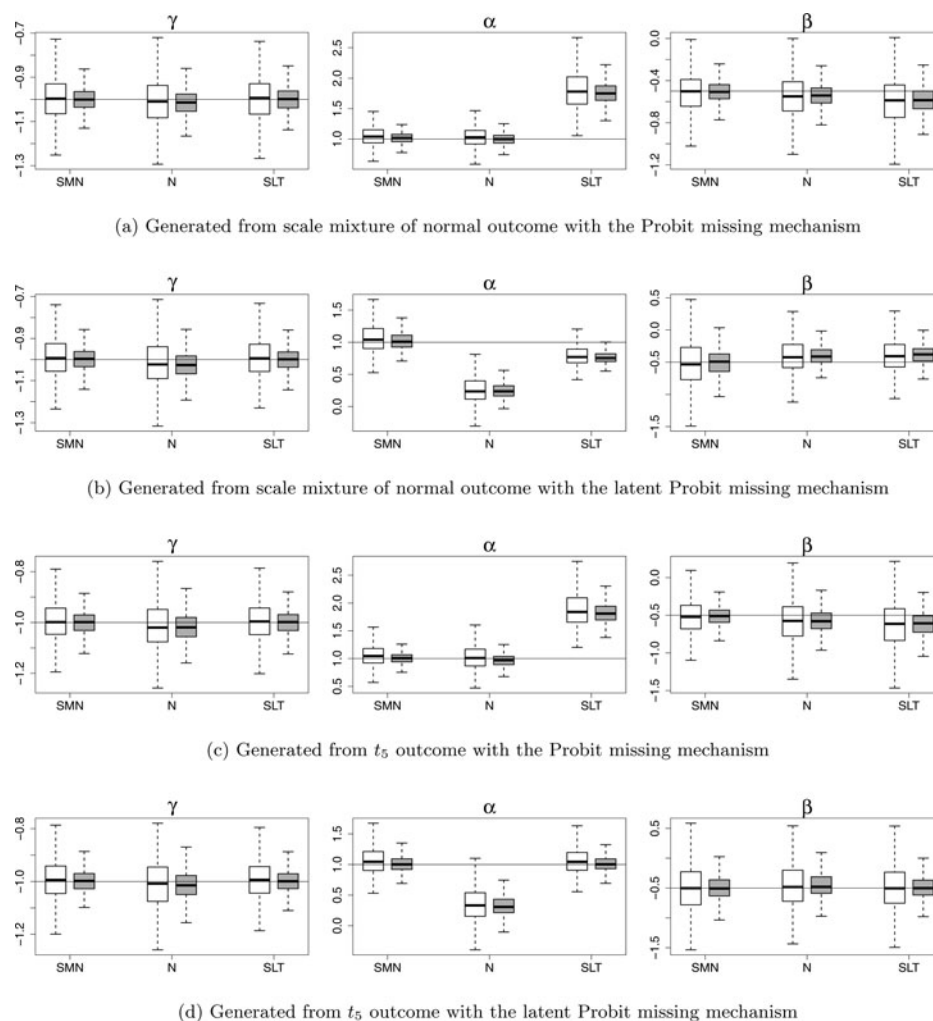


Figure 2. Estimates when $\beta = -0.5$. Data are generated from different outcome models, and analyzed under different outcome models with correct missing mechanisms: two-component scale mixture of normal outcome (SMN), normal outcome (N), and the selection- t model (SLT). The estimators of SLT are reparametrized to be equivalent to the scale mixture of normals with the latent Probit missing mechanism as shown in (8). In each boxplot, white boxes are for sample size 500 and gray boxes for 1500. The horizontal lines illustrate the true values of the parameters.

We fit the datasets by a two-component general mixture of normal outcome models with the Probit missing mechanism for cases 1 and 3, and a latent Probit missing mechanism for case 2. We also fit the datasets by the normal outcome with the Probit missing mechanism. Because the normal outcome model returns only one set of location parameters, we focus only on its estimators of missing mechanisms.

Figure 3 shows the simulation results for $\beta = -0.5$, and those for $\beta = -1$ are similar and omitted. To save space, we present only the results for the coefficients of X_1 and Y . Using normal mixture models for estimation, the biases for parameters ($\gamma_1, \theta, \alpha, \beta$) are small, and they become smaller as sample size increases. But the estimators by normal outcome model are very biased. Therefore, we recommend using a general mixture of normal models for outcomes with multimodality.

7. Application

We apply our models to the ambulatory expenditure data previously analyzed by Cameron and Trivedi (2009). The dataset contains the log of ambulatory expenditure (ambexp) and covariates $X = (1, \text{age}, \text{female}, \text{educ}, \text{blhisp}, \text{totchr},$

$\text{ins}, \text{income})$, including age, gender, education status, ethnicity, number of chronic diseases, insurance status, and income. The dataset contains 3328 observations and there are 526 missing values of ambexp. More details about the data can be found in chap. 16 of Cameron and Trivedi (2009). The authors applied the Heckman selection model to the data, and they found no significant selection effect.

However, we suspect that the samples do not exactly follow the normal model. We apply the general normal mixture model to the data and compare the results with those of the Heckman selection model and the selection- t model. We present the results in Table 1. We start with two components and observe that the standard deviations of the two components, σ_1 and σ_2 , are very different. The estimate of the ratio σ_1/σ_2 is 0.610 with 95% bootstrapping confidence interval (0.500, 0.716) excluding one. It provides strong empirical evidence of the existence of two latent normal components rather than a single one. We continue to fit a general normal mixture model with three components, and find that one component has estimated proportion 0.041. We not only find that the proportion of the third component is small, but also find that the regression coefficients of the third component are similar to one component. Therefore, we choose

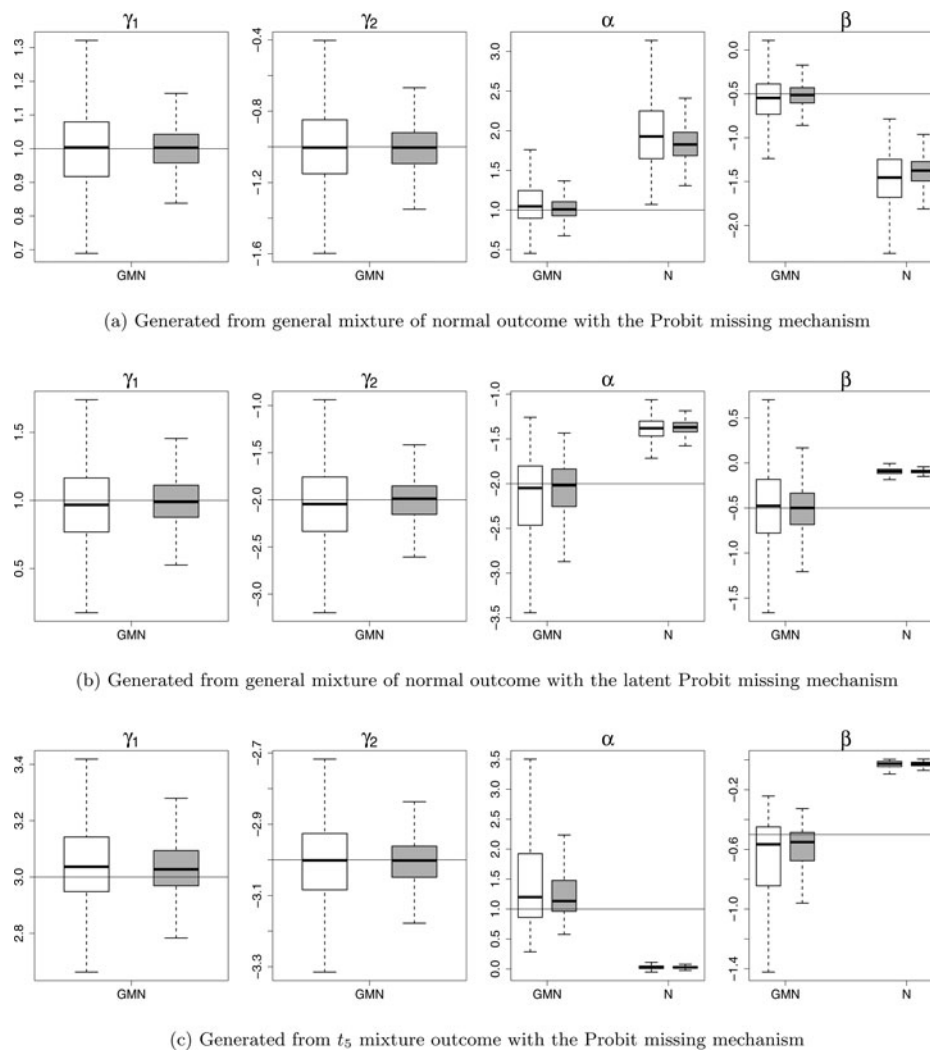


Figure 3. Estimates when $\beta = -0.5$. Data are generated from different outcome models, and analyzed under different outcome models with correct missing mechanisms: two-component general mixture of normal outcome (GMN), and normal outcome (N). In each boxplot, white boxes are for sample size 500 and gray boxes for 1500. The horizontal lines illustrate the true values of the parameters.

a general normal mixture model with two components, and omit analysis with more components.

The general normal mixture model with the Probit missing mechanism yields interesting practical interpretations. In our analysis, we first fit a general normal mixture model with different coefficients on all the covariates for the two components, and find that all regression coefficients are nearly identical except for *ins*. Therefore, in the last two columns of Table 1, we choose to present the results under a more parsimonious model allowing for only different regression coefficients of *ins*. The covariate *ins* has a positive although insignificant effect on $\log(\text{ambexp})$ in a large group of the people (88%) with an estimate 0.009 and 95% confidence interval $(-0.095, 0.174)$. However, *ins* has a significant negative effect in a small group (12%) with an estimate -0.595 and 95% confidence interval $(-1.705, -0.152)$. The results provide further evidence of two different components. The Heckman selection model or the selection- t model cannot detect such latent heterogeneity, because they are not able to accommodate location mixture or multimodality. As a result, these two models mix

the two groups, and consequently report no significant impact of *ins* on $\log(\text{ambexp})$. The two components in the normal mixture model may reflect different health statuses. For healthy people who rarely needed health care, the insurance had an insignificant effect on ambulatory expenditure; for sick people who needed more health care, the insurance was beneficial for reducing ambulatory expenditure. The two components might also reflect different types of health insurance. Sick people tended to buy insurance that had larger coverage proportions, and consequently the insurance helped reduce more ambulatory expenditure.

Previous analyses found evidence of heavy-tailedness in the outcome distribution. However, the upper confidence limit of ν in the selection- t model is near 20 (Marchenko and Genton 2012; Ding 2014), and such heavy-tailedness seems to be captured also by the normal mixture model as illustrated in the simulation studies in Sections 6.2 and 6.3. Furthermore, the Heckman selection model finds insignificant selection effect, but both the selection- t and the normal mixture model with the Probit missing mechanism find significant selection effect.

Table 1. The ambulatory expenditure study example.

	Selection		Selection-t		General normal mixture	
Outcome model						
age	0.212	(0.167, 0.257)	0.207	(0.163, 0.251)	0.209	(0.165, 0.253)
female	0.348	(0.230, 0.466)	0.307	(0.196, 0.417)	0.327	(0.213, 0.431)
educ	0.019	(−0.002, 0.039)	0.017	(−0.003, 0.037)	0.020	(0.000, 0.040)
blhisp	−0.219	(−0.336, −0.102)	−0.193	(−0.306, −0.080)	−0.211	(−0.326, −0.094)
totchr	0.540	(0.463, 0.617)	0.513	(0.443, 0.583)	0.529	(0.460, 0.588)
ins	−0.030	(−0.130, 0.070)	−0.053	(−0.151, 0.046)	0.009	(−0.095, 0.174)
					−0.595	(−1.705, −0.152)
σ	1.271	(1.236, 1.308)	1.195	(1.146, 1.246)	1.159	(0.958, 1.227)
					1.900	(1.423, 2.383)
Selection model						
income	0.003	(0.000, 0.005)	0.003	(0.000, 0.006)	0.003	(0.000, 0.006)
age	0.088	(0.034, 0.142)	0.099	(0.040, 0.157)	0.127	(0.060, 0.208)
female	0.663	(0.543, 0.782)	0.725	(0.591, 0.859)	0.734	(0.609, 0.911)
educ	0.062	(0.038, 0.086)	0.065	(0.040, 0.090)	0.066	(0.045, 0.091)
blhisp	−0.364	(−0.485, −0.243)	−0.394	(−0.524, −0.263)	−0.407	(−0.548, −0.291)
totchr	0.797	(0.658, 0.936)	0.890	(0.719, 1.061)	0.906	(0.750, 1.137)
ins	0.170	(0.047, 0.293)	0.180	(0.048, 0.313)	0.166	(0.047, 0.289)
ρ	−0.131	(−0.401; 0.161)	−0.322	(−0.526, −0.083)		
β_Y					−0.174	(−0.414, −0.025)
π_1	1		1		0.880	(0.401, 0.971)
log L	−5836.219		−5822.076			−5820.457
AIC	11706.44		11676.15			11680.91

NOTE: Point estimates and 95% confidence intervals of the Heckman selection model (columns 2 and 3), the selection-*t* model (columns 4 and 5), and the general mixtures of normal model (columns 6 and 7). The ρ parameter is only for the Heckman selection model and selection-*t* model, and the β_Y parameter is only for the general mixture of normal model. The π_1 parameter is the proportion of the first latent component, log *L* is the log-likelihood of the observed data, and AIC is the Akaike information criterion.

8. Discussion

Dealing with missing data is crucial for many applied problems, which are often challenging if the missing mechanisms are non-ignorable. Even if we have a fully parametric model on the missing mechanism, nonparametric outcome models subject to non-ignorable missing data are often not identifiable. In this article, we have demonstrated the identifiability of normal and normal mixture models with nonignorable missing data. The identifiability results do not require an instrumental variable for the missing mechanism, or the exclusion restriction assumption in the Heckman selection model (Wooldridge 2010), that is, there exists a covariate that is only in the selection equation but not in the outcome equation. Although the selection-*t* model could accommodate heavy-tailedness of the data, it could not model outcome with multimodality or latent groups. Our *t* mixture model with a nonignorable missing mechanism filled in the gap by allowing for modeling heavy-tailedness and multimodality simultaneously.

There are a few questions beyond the current scope of this article. First, determining the number of components in the mixture models (Lo, Mendell, and Rubin 2001; Chen, Li, and Fu 2012) is an important problem in practice. Second, although the *t* mixture model with the Robit missing mechanism, as discussed in Theorem 5, is useful for modeling heavy-tailed data, its maximum likelihood estimates involve numerical integrations. We are going to develop an efficient Bayesian inferential procedure for it.

Supplementary Materials

Supplementary materials contain the proofs of the theorems and corollaries.

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